

# Exceptional-Set Transfer for Short-Shift Factor-Ray Energies: The Two-Fifteenths Window, Parity Conditioning, and Fixed-Ray Obstructions

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## Abstract

Put  $u(n) = \Lambda(n) - 1$  and let  $W \in C_c^1((0, \infty); \mathbb{R})$ . For a fixed integer shift  $h$  consider

$$\mathcal{R}_{h,D}(X; t) = \sum_{\substack{n > D \\ m \geq 1}} u(n) (\Lambda(mn + h) - 1) W(mn/X) (n/m)^{it}.$$

Writing  $n = ak$ ,  $m = bk$ ,  $(a, b) = 1$ , compresses equal Mellin frequencies into primitive factor rays and defines the ray energy  $\mathcal{E}_{h,D}(X)$  as the sum of the squared compressed coefficients. Earlier long-shift arguments evaluate this energy after averaging over  $h$  on a scale  $H \geq X$ . We lower that scale unconditionally.

The new input is a weighted exceptional-set transfer. Guth and Maynard proved a prime number theorem in intervals of length  $H \geq X^{2/15+\varepsilon}$  for all but a stretched-exponentially sparse set of integer centers. The diagonal factor-product center  $r = mn$  carries the weight

$$A_D(r) = |W(r/X)|^2 \sum_{\substack{n|r \\ n > D}} (\Lambda(n) - 1)^2.$$

We prove  $\sum_{r \lesssim X} A_D(r)^2 \ll X(\log X)^4$ ; hence every such exceptional set has  $A_D$ -mass smaller than  $X(\log X)^{-A}$  for every  $A > 0$ . This rules out concentration of the short-interval exceptions on the factor products relevant to the ray energy.

Let  $G \in C_c^1((0, \infty))$  be nonnegative and nonzero. Fix  $0 < \varepsilon < 13/15$  and  $0 < \eta < 1$ . Uniformly for

$$X^{2/15+\varepsilon} \leq H \leq X, \quad 1 \leq D \leq X^{1-\eta},$$

we prove

$$\begin{aligned} \sum_h G(h/H) \mathcal{E}_{h,D}(X) &= \frac{1}{2} H X I_0(G) I_2(W) \log X \\ &\quad \times ((\log X)^2 - (\log D)^2) \\ &\quad + O(H X (\log X)^2 \log \log(3X)). \end{aligned}$$

The same formula holds for the raw target energy. The diagonal target square supplies  $\log X$ ; all same-ray off-diagonals use only a two-linear-forms Selberg upper sieve. The first-moment model and the hard- and Fejér-band consequences extend to the same shift range. A parity-conditioned even-shift ensemble has the corresponding coefficient  $1/4$  and still selects no prescribed even shift.

We also prove exact fixed-shift boundaries. For nonnegative, nonzero  $W$ , at  $h = 2$  an explicit family of primitive rays (for example, those with  $24 \mid a$ ,  $ab = o(X)$ , and the stated inactive cutoff) is sign coherent and has coefficient of natural radial size, so uniform individual-ray cancellation is false. On the locally admissible ray  $(a, b, h) = (6, 1, 1)$ , coefficient evaluation is algebraically equivalent to a weighted prime number theorem for  $6k^2 + 1$ . Finally, an exact minimax formula shows that one Mellin coordinate observes only the sum of the radial coefficients on a ray. These results explain why the growing-window theorem cannot be specialized to  $H = O(1)$ . No fixed-shift prime-pair asymptotic, twin-prime lower bound, quadratic-prime theorem, new distribution level, or breach of the parity barrier is claimed.

**Keywords:** primes in almost all short intervals; exceptional-set transfer; factor-ray energy; Mellin spectrum; Selberg upper-bound sieve; sieve parity.

**MSC 2020:** Primary 11N05, 11N13, 11N36; Secondary 11M06, 42A05.

## 1 Introduction and scope

The factor-ray transform separates two different kinds of averaging. Continuous Mellin averaging separates distinct reduced ratios  $a/b$ , whereas arithmetic averaging of the shift  $h$  controls the prime targets that remain coherent on a single ray. TPC-11 established the first separation at fixed shift and evaluated the resulting ray energy after a smooth shift average of length  $H \geq X$  [12]. Its explicit next question was whether one can shorten the shift average without pretending that a fixed shift has been reached.

The recent large-values theorem of Guth and Maynard [5] changes the answer. Among its consequences is a prime number theorem in intervals of length  $X^{2/15+\varepsilon}$  for all but a stretched-exponentially sparse set of integer starting points. The exponent  $2/15$  applies to almost all intervals; it is logically different from the exponent  $17/30$  in their uniform short-interval theorem. The exceptional-set framework and its current quantitative refinements are also discussed by Gafni and Tao [4]. We use the almost-all result because the ray energy already sums over the product center  $r = mn$ .

That last sentence contains the main technical issue. A set that is sparse among all integers could, in principle, carry most of a structured divisor weight. The relevant weight here is not bounded:

$$A_D(r) = |W(r/X)|^2 \sum_{\substack{n|r \\ n>D}} (\Lambda(n) - 1)^2.$$

We close this gap by proving a global second-moment bound for  $A_D$  and then transferring the exceptional-set estimate by Cauchy–Schwarz. This weighted transfer, rather than a pointwise short-interval theorem, is what allows the shift scale to fall below the uniform threshold.

### 1.1 Main results

Throughout the paper

$$u(n) = \Lambda(n) - 1, \quad \eta(q) = \Lambda(q) - 1,$$

and  $W \in C_c^1((0, \infty); \mathbb{R})$  has support in a fixed interval  $[\alpha, \beta] \subset (0, \infty)$ . We write

$$\mathcal{T}_{h,D}(X; t) = \sum_{\substack{n>D \\ m \geq 1}} u(n) \Lambda(mn + h) W(mn/X) (n/m)^{it}, \quad (1)$$

$$\mathcal{M}_D(X; t) = \sum_{\substack{n>D \\ m \geq 1}} u(n) W(mn/X) (n/m)^{it}, \quad \mathcal{R}_{h,D} = \mathcal{T}_{h,D} - \mathcal{M}_D. \quad (2)$$

The factorization  $n = ak$ ,  $m = bk$ ,  $(a, b) = 1$ , gives

$$\mathcal{R}_{h,D}(X; t) = \sum_{(a,b)=1} c_{a,b}(h; X, D) (a/b)^{it}, \quad (3)$$

where

$$c_{a,b}(h; X, D) = \sum_{\substack{k \geq 1 \\ ak > D}} u(ak) \eta(abk^2 + h) W(abk^2/X). \quad (4)$$

The centered ray energy is

$$\mathcal{E}_{h,D}(X) = \sum_{(a,b)=1} |c_{a,b}(h; X, D)|^2. \quad (5)$$

Our first theorem transfers the almost-all short-interval prime number theorem through both the signed first-moment coefficients and the positive diagonal source weights. In particular, for every  $A > 0$ ,

$$\frac{1}{H} \sum_h G(h/H) \mathcal{T}_{h,D}(X; t) = I_0(G) \mathcal{M}_D(X; t) + O_A(X(\log X)^{-A}) \quad (6)$$

uniformly in  $t \in \mathbb{R}$  throughout the growing parameter range of the energy theorem below.

The main energy theorem is the following. Put

$$I_0(G) = \int_0^\infty G(v) dv, \quad I_2(W) = \int_0^\infty |W(v)|^2 dv. \quad (7)$$

For fixed  $0 < \varepsilon < 13/15$  and  $0 < \eta < 1$ , uniformly in

$$X^{2/15+\varepsilon} \leq H \leq X, \quad 1 \leq D \leq X^{1-\eta},$$

we prove

$$\begin{aligned} \sum_h G(h/H) \mathcal{E}_{h,D}(X) &= \frac{1}{2} H X I_0(G) I_2(W) \log X \\ &\quad \times ((\log X)^2 - (\log D)^2) \\ &\quad + O_{\varepsilon, \eta, W, G}(H X (\log X)^2 \log \log(3X)). \end{aligned} \quad (8)$$

The raw ray energy has the same asymptotic. In this at-most-linear shift range the correct target logarithm is  $\log X$ , not  $\log H$ .

The source diagonal is of order  $X(\log X)^2$ . On a nonexceptional center, the target square contributes  $H \log X$ . A two-linear-forms Selberg upper sieve bounds each same-ray off-diagonal by  $H \log \log X$ , after which the source off-diagonal ledger from TPC-11 is  $O(X \log^2 X)$ . Thus the extra logarithm on the diagonal remains the sole source of the asymptotic. The proof does not obtain cancellation in an individual fixed-shift quadratic correlation.

The exact factor-ray mean-square identities then convert (8) into hard- and Fejér-band results over the same short shift window. We also prove a parity-conditioned version over even shifts. Its main coefficient is  $1/4$  rather than  $1/2$ , and it does not select any prescribed even shift.

## 1.2 The fixed-shift boundary

Shrinking a growing average and setting  $H = 1$  are not interchangeable operations. We make that failure concrete in three ways.

- (i) For nonnegative, nonzero  $W$  and every fixed  $h_0 \geq 2$ , there is an infinite family of primitive rays on which the actual arithmetic summands in (4) have one sign and no cancellation at all. This includes balanced ray sectors whenever the cutoff is below the square-root scale; a separate unbalanced construction covers the full range  $D \leq X^{1-\eta}$ .

- (ii) On the locally admissible ray  $(a, b, h) = (6, 1, 1)$ , the unknown part of the coefficient is exactly

$$\sum_k \Lambda(6k^2 + 1)W(6k^2/X).$$

Therefore a natural-scale coefficient asymptotic is equivalent to the corresponding weighted quadratic-prime asymptotic. Bateman–Horn predicts such an asymptotic [3], but it is not known; the almost-prime theorem for  $n^2 + 1$  does not supply it [8].

- (iii) The whole one-Mellin signal on one ray is a known exponential times the single scalar  $\sum_k x_k$ . We give the exact minimax error for recovering any other radial functional from that scalar. This is an information statement for coefficient-blind recovery, not a prohibition on using additional arithmetic input.

These statements identify both a local obstruction and an admissible open problem before the endpoint prime-pair layer is reached. At the endpoint, the selector in TPC-11 already shows that a sufficiently strong fixed-shift band approximation would recover a weighted Hardy–Littlewood functional [7, 12].

### 1.3 Claim ledger

| Statement                                          | Status here                                   | What it does not imply                        |
|----------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| Weighted transfer of almost-all short-interval PNT | Proved in the displayed growing range         | No prescribed shift is controlled             |
| Short-shift factor-ray energy asymptotic           | Proved with explicit error                    | No individual ray or target pair is evaluated |
| Even-shift ensemble                                | Proved after parity centering                 | No selection of a particular even shift       |
| Fixed-ray cancellation                             | False on an explicit local-obstruction family | Does not contradict averaged energy           |
| Admissible quadratic ray                           | Exact algebraic reduction                     | The quadratic-prime asymptotic remains open   |

No twin-prime lower bound, fixed-shift Hardy–Littlewood asymptotic, quadratic-prime theorem, new level of distribution, or breach of the sieve parity barrier is asserted.

## 2 Factor-ray setup and source ledgers

Fix numbers  $0 < \alpha < \beta$  such that  $\text{supp } W \subset [\alpha, \beta]$ . The shift is an integer, and  $X$  is always large enough that every target selected by the weights is positive. All sums are finite.

### 2.1 Ray compression and the inherited spectral identities

For every factor pair, write uniquely

$$n = ak, \quad m = bk, \quad (a, b) = 1, \quad k = (m, n). \quad (9)$$

Then  $mn = abk^2$  and  $n/m = a/b$ , so substitution gives (3)–(4). The raw ray coefficient is

$$c_{a,b}^{\text{raw}}(h; X, D) = \sum_{\substack{k \geq 1 \\ ak > D}} u(ak) \Lambda(abk^2 + h) W(abk^2/X), \quad (10)$$

and its squared energy is denoted by  $\mathcal{E}_{h,D}^{\text{raw}}(X)$ .

Two geometric facts from TPC-11 will be used below. We record them to fix all normalizations. Distinct primitive rays in the support satisfy

$$\left| \log \frac{a}{b} - \log \frac{c}{d} \right| \geq \delta_X, \quad \delta_X = 2 \operatorname{arsinh} \left( \frac{1}{2\beta X} \right). \quad (11)$$

Indeed,  $|ad - bc| \geq 1$  and  $abcd \leq (\beta X)^2$ ; the identity

$$2 \sinh \left( \frac{1}{2} \left| \log \frac{ad}{bc} \right| \right) = \frac{|ad - bc|}{\sqrt{abcd}}$$

proves the claim.

The generalized Hilbert inequality of Montgomery and Vaughan [11] therefore gives, for every bounded interval  $I \subset \mathbb{R}$ ,

$$\int_I |\mathcal{R}_{h,D}(X; t)|^2 dt = |I| \mathcal{E}_{h,D}(X) + O_W(X \mathcal{E}_{h,D}(X)). \quad (12)$$

The same identity holds for the raw transform. With the Fourier convention  $\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-it\xi} dt$  and

$$\kappa(t) = \frac{2}{\pi} \left( \frac{\sin(t/2)}{t} \right)^2, \quad \widehat{\kappa}(\xi) = (1 - |\xi|)_+, \quad (13)$$

Fourier expansion gives the exact formula

$$\int_{\mathbb{R}} \kappa(t/T) |\mathcal{R}_{h,D}(X; t)|^2 dt = T \mathcal{E}_{h,D}(X) \quad \text{whenever } T\delta_X \geq 1. \quad (14)$$

The endpoint value of  $\widehat{\kappa}$  is zero, so equality is valid also when a scaled gap equals one.

## 2.2 Weights on product centers

For an integer  $r \geq 1$ , define

$$A_D(r) := |W(r/X)|^2 \sum_{\substack{n|r \\ n>D}} u(n)^2, \quad (15)$$

$$B_{D,t}(r) := W(r/X) \sum_{\substack{n|r \\ n>D}} u(n) (n^2/r)^{it}. \quad (16)$$

The first is the source weight on the target-square diagonal. The second is the coefficient that appears after the first-moment transform is grouped by  $r = mn$ .

**Lemma 2.1** (Pointwise and quadratic product ledgers). *Uniformly in  $D \geq 1$  and  $t \in \mathbb{R}$ ,*

$$A_D(r) \ll_W \mathbf{1}_{[\alpha X, \beta X]}(r) ((\log(2r))^2 + \tau(r)), \quad (17)$$

$$|B_{D,t}(r)| \ll_W \mathbf{1}_{[\alpha X, \beta X]}(r) (\log(2r) + \tau(r)), \quad (18)$$

and

$$\sum_{r \geq 1} A_D(r)^2 \ll_W X (\log(2X))^4, \quad (19)$$

$$\sum_{r \geq 1} |B_{D,t}(r)|^2 \ll_W X (\log(2X))^3. \quad (20)$$

*Proof.* Since  $(\Lambda - 1)^2 \leq 2(1 + \Lambda^2)$ ,

$$\sum_{d|r} u(d)^2 \leq 2\tau(r) + 2 \sum_{p^j|r} (\log p)^2.$$

If  $r = \prod_p p^{v_p}$ , then

$$\sum_{p^j|r} (\log p)^2 = \sum_p v_p (\log p)^2 \leq \left( \sum_p v_p \log p \right)^2 = (\log r)^2.$$

This proves (17). Also

$$\sum_{d|r} |u(d)| \leq \sum_{d|r} \Lambda(d) + \tau(r) = \log r + \tau(r),$$

which proves (18), uniformly because the phase has modulus one. Finally use the classical estimate  $\sum_{r \leq y} \tau(r)^2 \ll y(\log(2y))^3$  and the fixed support of  $W$ .  $\square$

The total mass of  $A_D$  has an asymptotic that will provide the main coefficient.

**Lemma 2.2** (Source diagonal). *For every fixed  $\varepsilon > 0$ , uniformly for  $1 \leq D \leq X^{1-\varepsilon}$ ,*

$$\begin{aligned} \mathcal{S}_D(X) &:= \sum_{r \geq 1} A_D(r) = \sum_{\substack{n > D \\ m \geq 1}} u(n)^2 |W(mn/X)|^2 \\ &= \frac{1}{2} X I_2(W) ((\log X)^2 - (\log D)^2) + O_W(X \log(2X)). \end{aligned} \tag{21}$$

*Proof.* Euler summation gives

$$\sum_{m \geq 1} |W(mn/X)|^2 = \frac{X}{n} I_2(W) + O_W(1).$$

The classical zero-free-region prime number theorem and partial summation give [9, Chapter 5]

$$\sum_{n \leq y} \frac{(\Lambda(n) - 1)^2}{n} = \frac{1}{2} (\log y)^2 - \log y + O(1), \tag{22}$$

while  $\sum_{n \leq y} u(n)^2 \ll y \log(2y)$ . Insert the Euler estimate, subtract the range  $n \leq D$ , and absorb the fixed support endpoints into  $O_W(X \log X)$ .  $\square$

### 2.3 The same-ray off-diagonal ledger

For later use put

$$\begin{aligned} \mathcal{L}_D(X) &:= \sum_{(a,b)=1} \sum_{\substack{k \neq \ell \\ ak, a\ell > D}} |u(ak)u(a\ell)| \\ &\quad \times |W(abk^2/X)W(ab\ell^2/X)|. \end{aligned} \tag{23}$$

**Lemma 2.3** (Source-ray square and off-diagonal ledgers). *Uniformly for  $Y \geq 2$ ,*

$$\sum_{ak^2 \leq Y} \frac{u(ak)^2}{ak} \ll (\log(2Y))^2, \tag{24}$$

and, for every  $D \geq 1$ ,

$$\mathcal{L}_D(X) \ll_W X(\log(2X))^2. \tag{25}$$

*Proof.* For (24), the contribution of the constant part of  $u^2 \leq 2(1 + \Lambda^2)$  is

$$\sum_{k \leq \sqrt{Y}} \frac{1}{k} \sum_{a \leq Y/k^2} \frac{1}{a} \ll (\log(2Y))^2.$$

If  $ak = p^j$ , then  $k = p^\ell$  and  $a = p^{j-\ell}$ . Dropping the extra constraint  $p^{j+\ell} \leq Y$  gives

$$\sum_{p^j \leq Y} (j+1) \frac{(\log p)^2}{p^j} \ll (\log(2Y))^2.$$

For the second assertion, let  $K_{a,b}$  be the radial indices selected by the two weights and put  $x_k = |u(ak)W(akk^2/X)|$ . Their fixed multiplicative support implies  $|K_{a,b}| \ll_W k$  for every  $k \in K_{a,b}$ . Hence

$$\sum_{k \neq \ell} x_k x_\ell \leq |K_{a,b}| \sum_k x_k^2 \ll_W \sum_{k \in K_{a,b}} k x_k^2.$$

For fixed  $a, k$ , the number of supported  $b$  is  $O_W(X/(ak^2))$ ; the additive one is absorbed by support nonemptiness. Summing first over  $b$  gives

$$\mathcal{L}_D(X) \ll_W X \sum_{ak^2 \ll_W X} \frac{u(ak)^2}{ak},$$

and (24) completes the proof.  $\square$

### 3 Exceptional-set transfer at the two-fifteenths scale

This section isolates the new analytic interface. We first state the short-interval input in the precise form from which we need it.

**Theorem 3.1** (Guth–Maynard short-interval inputs). *Fix  $\rho > 0$ . If  $Y \in [X^{2/15+\rho}, X^{0.99}]$ , then all but (a set that may depend on  $Y$ )*

$$O_\rho\left(X \exp\{-c_\rho(\log X)^{1/4}\}\right) \tag{26}$$

*integers  $x \in [X, 2X]$  satisfy*

$$\pi(x+Y) - \pi(x) = \frac{Y}{\log x} + O_\rho\left(Y \exp\{-c_\rho(\log X)^{1/4}\}\right). \tag{27}$$

*If instead*

$$Y \in [X^{17/30+\rho}, X^{0.99}],$$

*then (27) holds for every integer  $x \in [X, 2X]$ .*

These are Corollaries 1.3 and 1.4 of arXiv:2405.20552v2 [5], after harmless changes in the positive constant in the stretched exponential. The almost-all theorem is stated for integer centers, which is exactly what is needed here.

#### 3.1 Smooth target moments outside a sparse set

Let  $G \in C_c^1((0, \infty))$  be fixed. Choose  $0 < \gamma < \Gamma$  with  $\text{supp } G \subset [\gamma, \Gamma]$ .

**Lemma 3.2** (Smooth translated moments). *Fix  $0 < \varepsilon < 13/15$ . Uniformly for*

$$X^{2/15+\varepsilon} \leq H \leq X, \quad (28)$$

*there is a set of integers  $\mathfrak{E} = \mathfrak{E}(X, H, G) \subset [\alpha X, \beta X]$  with*

$$|\mathfrak{E}| \ll_{\varepsilon, W, G} X \exp\{-c_{\varepsilon, G}(\log X)^{1/4}\} \quad (29)$$

*such that every  $r \in [\alpha X, \beta X] \setminus \mathfrak{E}$  satisfies, for every  $A > 0$ ,*

$$\sum_h G(h/H) \Lambda(r+h) = HI_0(G) + O_{A, \varepsilon, G}(H(\log X)^{-A}), \quad (30)$$

$$\sum_h G(h/H) \Lambda(r+h)^2 = HI_0(G) \log X + O_{\varepsilon, G}(H), \quad (31)$$

$$\sum_h G(h/H) (\Lambda(r+h) - 1)^2 = HI_0(G) \log X + O_{\varepsilon, G}(H). \quad (32)$$

*Proof.* It is enough to treat  $r$  in one of finitely many dyadic intervals contained in  $[\alpha X, \beta X]$ . We separate two ranges.

First suppose  $H \leq X^{0.98}$ . Approximate  $G$  by a step function on

$$L = \left\lceil \exp\{(\log X)^{1/8}\} \right\rceil$$

equal subintervals of  $[\gamma, \Gamma]$ . The prime count on each step is a difference of two cumulative counts with lengths  $y_j H$ , where  $\gamma \leq y_j \leq \Gamma$ . Apply the almost-all part of [theorem 3.1](#), with its parameter smaller than  $\varepsilon$ , to these  $O(L)$  lengths and take the union of their exceptional sets. For large  $X$ , the fixed endpoint factors put all these lengths between  $X^{2/15+\varepsilon/2}$  and  $X^{0.99}$ . Since  $\log L = (\log X)^{1/8} = o((\log X)^{1/4})$ , the union does not change the shape of (29). Partial summation gives the prime part of (30); the step-approximation error is  $O(H/L)$  and is therefore smaller than  $H(\log X)^{-A}$  for every fixed  $A$ , using one exceptional set independent of  $A$ .

Now suppose  $H > X^{0.98}$ . Put  $N = \lfloor (\Gamma - \gamma)H/X^{0.97} \rfloor$  and partition  $[\gamma H, \Gamma H]$  into  $N$  equal consecutive blocks. For large  $X$ , every block length lies between  $X^{0.97}$  and  $2X^{0.97}$ , so there is no short remainder; changing integer endpoints by  $O(1)$  is harmless. Since  $H \leq X$ , every block starts at height comparable to  $X$ . The all-interval part of [theorem 3.1](#), with any fixed small value of  $\rho$ , applies to every block for large  $X$ . Freeze  $G(h/H)$  on each block. The total error from this freezing is  $O_G(X^{0.97}) = O_{A, G}(H(\log X)^{-A})$  for every fixed  $A$ , while the short-interval errors sum to  $O(H \exp\{-c(\log X)^{1/4}\})$ . Thus the prime part of (30) holds at every center in this range; we may take  $\mathfrak{E} = \emptyset$ .

In both ranges, the omitted proper prime powers contribute at most

$$\ll_G \left( HX^{-1/2} + \log(2X) \right) \log(2X) = o_{A, \varepsilon}(H(\log X)^{-A})$$

for every fixed  $A$ . This completes the proof of (30) for the full von Mangoldt function.

For the square, first restrict to prime target values. On the support,  $r + h \asymp X$ , and

$$\Lambda(p)^2 = (\log p)^2 = \log X \Lambda(p) + O_G(\Lambda(p)).$$

Equations (30) and partial summation therefore give the main term in (31), with an  $O(H)$  remainder. Proper prime powers in an interval of length  $O(H)$  at height  $X$  contribute

$$\ll_G \left( HX^{-1/2} + \log(2X) \right) (\log(2X))^2 = o_{\varepsilon}(H).$$

Indeed, for each exponent  $j \geq 2$ , consecutive  $j$ th powers at height  $X$  are separated by  $\gg X^{1-1/j}$ , and there are  $O(\log X)$  possible exponents. This proves (31). Finally expand  $(\Lambda - 1)^2 = \Lambda^2 - 2\Lambda + 1$  and use  $\sum_h G(h/H) = HI_0(G) + O_G(1)$ .  $\square$

### 3.2 The weighted transfer

The preceding exceptional set could depend on  $H$  and on the step approximation. Its exact location is immaterial because its mass under the factor-product weights is negligible.

**Proposition 3.3** (Weighted exceptional-set transfer). *Let  $\mathfrak{E} \subset [\alpha X, \beta X] \cap \mathbb{Z}$  satisfy*

$$|\mathfrak{E}| \ll X \exp\{-c(\log X)^{1/4}\} \quad (33)$$

*for some fixed  $c > 0$ . Then, uniformly in  $D \geq 1$  and  $t \in \mathbb{R}$ , for every  $A > 0$ ,*

$$\sum_{r \in \mathfrak{E}} A_D(r) \ll_{A,c,W} X(\log X)^{-A}, \quad (34)$$

$$\sum_{r \in \mathfrak{E}} |B_{D,t}(r)| \ll_{A,c,W} X(\log X)^{-A}. \quad (35)$$

*Proof.* Cauchy–Schwarz and lemma 2.1 give

$$\sum_{r \in \mathfrak{E}} A_D(r) \leq |\mathfrak{E}|^{1/2} \left( \sum_r A_D(r)^2 \right)^{1/2} \ll_W X(\log X)^2 e^{-c(\log X)^{1/4}/2}.$$

The stretched exponential absorbs every fixed power of  $\log X$ . The same argument, using (20), proves (35).  $\square$

*Remark 3.4* (What is new and what is imported). The prime number theorem in almost all short intervals is the deep input of Guth and Maynard [5]. The step needed in the present factor-ray setting is proposition 3.3: it verifies that their exceptional centers cannot concentrate on the product positions with the unbounded source weight generated by the factor-ray diagonal. We do not claim a new short-interval prime number theorem.

## 4 Short-shift moments after weighted transfer

For the remainder of the positive theory, fix numbers  $0 < \varepsilon < 13/15$  and  $0 < \eta < 1$ . Thus

$$\frac{2}{15} + \varepsilon < 1, \quad (36)$$

and assume

$$X^{2/15+\varepsilon} \leq H \leq X, \quad 1 \leq D \leq X^{1-\eta}. \quad (37)$$

The support of  $G$  is contained in  $(0, \infty)$ , so the average samples a growing interval  $h \asymp_G H$ ; it does not contain any fixed shift once  $X$  is large.

### 4.1 The first-moment model

**Theorem 4.1** (Short-shift first moment). *For every  $A > 0$ , uniformly for (37) and all  $t \in \mathbb{R}$ ,*

$$\frac{1}{H} \sum_h G(h/H) \mathcal{T}_{h,D}(X; t) = I_0(G) \mathcal{M}_D(X; t) + O_{A,\varepsilon,W,G}(X(\log X)^{-A}). \quad (38)$$

*Equivalently, the normalized smooth average of  $\mathcal{R}_{h,D}$  has the displayed error.*

*Proof.* Group the residual average by  $r = mn$ . Equations (2) and (16) give

$$\frac{1}{H} \sum_h G(h/H) \mathcal{R}_{h,D}(X; t) = \sum_r B_{D,t}(r) \left\{ \frac{1}{H} \sum_h G(h/H) (\Lambda(r+h) - 1) \right\}. \quad (39)$$

On the complement of the exceptional set in lemma 3.2, the braces are  $O_A((\log X)^{-A})$ . The pointwise estimate (18) and the classical mean order of  $\tau$  give

$$\sum_r |B_{D,t}(r)| \ll_W X \log(2X).$$

Request two additional logarithmic powers.

On the exceptional set, the braces are  $O_G(\log(2X))$  by the elementary upper-bound sieve for primes in an interval; the completely trivial  $O_G((\log X)^2)$  would also suffice. Apply (35) with a larger value of  $A$ . Both parts are uniform in  $t$  and show that the normalized average of  $\mathcal{R}_{h,D}$  is  $O_A(X(\log X)^{-A})$ . Finally,

$$\frac{1}{H} \sum_h G(h/H) = I_0(G) + O_G(H^{-1})$$

and the elementary bound  $|\mathcal{M}_D(X; t)| \ll_W X \log(2X)$  shows that the Riemann-sum error is absorbed by the stated remainder, since  $H \geq X^{2/15+\varepsilon}$ . This proves the displayed formula for  $\mathcal{T}_{h,D}$ .  $\square$

## 4.2 The transferred target square

**Proposition 4.2** (Weighted target-square transfer). *Uniformly in (37),*

$$\begin{aligned} \sum_{r \geq 1} A_D(r) \sum_h G(h/H) (\Lambda(r+h) - 1)^2 \\ = HI_0(G) \log X \mathcal{S}_D(X) + O_{\varepsilon, W, G}(HX(\log X)^2). \end{aligned} \quad (40)$$

The same formula holds with the target square replaced by  $\Lambda(r+h)^2$ .

*Proof.* For  $r$  outside the exceptional set, use (32); the total of its  $O(H)$  error is  $O(H\mathcal{S}_D(X)) = O(HX(\log X)^2)$  by lemma 2.2.

On the exceptional set, the whole inner target-square sum is at most  $O_G(H(\log(2X))^2)$ . The same bound covers the main term being removed there. Proposition 3.3, with an arbitrarily large logarithmic saving, makes the exceptional contribution  $O(HX(\log X)^{-10})$ , say. This proves the centered formula. Equation (31) proves the raw version in exactly the same way.  $\square$

The proposition is the point where the almost-all theorem becomes a complete asymptotic input: no product center is discarded from the final sum.

## 4.3 A uniform two-target upper sieve

For  $q \neq 0$ , put

$$\mathfrak{S}^+(q) = \prod_{\substack{p|q \\ p > 2}} \frac{p-1}{p-2}. \quad (41)$$

**Lemma 4.3** (Two translated targets in a short shift window). *Let  $r \neq r'$  lie in  $[\alpha X, \beta X]$ . Uniformly for  $H \geq X^{2/15+\varepsilon}$  and  $H \leq X$ ,*

$$\begin{aligned} \sum_h G(h/H) |(\Lambda(r+h) - 1)(\Lambda(r'+h) - 1)| \\ \ll_{\varepsilon, W, G} H \mathfrak{S}^+(r - r') \ll_{\varepsilon, W, G} H \log \log(3X). \end{aligned} \quad (42)$$

The first bound remains valid if one or both centered factors are replaced by  $\Lambda$ .

*Proof.* The Selberg upper-bound sieve for the two linear forms  $h + r$  and  $h + r'$  gives

$$\#\{h \asymp_G H : h + r, h + r' \text{ are prime}\} \ll_G \frac{H}{(\log H)^2} \mathfrak{S}^+(r - r').$$

Restoring the von Mangoldt weights costs  $(\log X)^2$ . Since  $H \geq X^{2/15+\varepsilon}$ , the ratio  $\log X / \log H$  is bounded in terms of  $\varepsilon$ , and the prime-prime contribution has the asserted size. The constant and one-prime pieces follow from the corresponding one-linear- form upper sieve.

If one target is a proper prime power, the local spacing argument used in [lemma 3.2](#) gives only a polylogarithmic term plus  $O(HX^{-1/2}(\log X)^2)$ , both  $o_\varepsilon(H)$ . Finally the standard maximal-order estimate

$$\mathfrak{S}^+(q) \ll \log \log(3|q|)$$

follows by comparing with the product over the smallest primes. Here  $0 < |r - r'| \ll_W X$ . See Halberstam and Richert [[6](#), Chapter 6] for the upper sieve.  $\square$

## 5 Short-shift ray energy and Mellin-band consequences

We now combine the transferred diagonal, the two-target upper sieve, and the source ledgers.

**Theorem 5.1** (Short-shift factor-ray energy). *Let  $G \in C_c^1((0, \infty))$  be nonnegative and nonzero. Under [\(36\)](#)–[\(37\)](#),*

$$\begin{aligned} \sum_h G(h/H) \mathcal{E}_{h,D}(X) &= \frac{1}{2} H X I_0(G) I_2(W) \log X \\ &\quad \times ((\log X)^2 - (\log D)^2) \\ &\quad + O_{\varepsilon, \eta, W, G}(H X (\log X)^2 \log \log(3X)). \end{aligned} \tag{43}$$

*The same asymptotic holds for the raw energy  $\mathcal{E}_{h,D}^{\text{raw}}(X)$ .*

*Proof.* Expand the square in [\(5\)](#) and interchange the finite sums. On the radial diagonal  $k = \ell$ , the target position is  $r = abk^2$ . The unique substitution  $n = ak$ ,  $m = bk$  identifies the diagonal contribution with the left-hand side of [\(40\)](#). Proposition [4.2](#) and [lemma 2.2](#) give

$$\begin{aligned} H I_0(G) \log X &\left\{ \frac{1}{2} X I_2(W) ((\log X)^2 - (\log D)^2) + O_W(X \log X) \right\} \\ &\quad + O_{\varepsilon, W, G}(H X (\log X)^2). \end{aligned}$$

This is the displayed main term with an admissible diagonal error.

For  $k \neq \ell$ , the target positions  $abk^2$  and  $abl^2$  are distinct. Apply [lemma 4.3](#), take absolute values of the source coefficients, and use [\(25\)](#). The whole off-diagonal is

$$\ll_{\varepsilon, W, G} H \log \log(3X) \mathcal{L}_D(X) \ll_{\varepsilon, W, G} H X (\log X)^2 \log \log(3X).$$

This proves the centered theorem. The raw diagonal is covered by the raw part of [proposition 4.2](#), and the same upper-sieve ledger handles its off-diagonal.  $\square$

If  $W \neq 0$ , then  $I_2(W) > 0$ . Because  $D \leq X^{1-\eta}$ ,

$$(\log X)^2 - (\log D)^2 \geq (2\eta - \eta^2)(\log X)^2. \tag{44}$$

Thus, for  $W \neq 0$ , the error in [\(43\)](#) is smaller than the main term by  $O_\eta(\log \log X / \log X)$ .

*Remark 5.2* (The exact advance over the long-shift theorem). TPC-11 used a prime number theorem at every translated center and hence assumed  $H \geq X$ . Here the diagonal is summed over all product centers, so the exceptional-set transfer covers  $X^{2/15+\varepsilon} \leq H \leq X$ . No cancellation is claimed in one fixed-shift same-ray correlation. The result is also distinct from the averaged quadratic-polynomial theorem of Baier and Zhao [2]: their averaging parameter is the additive coefficient of  $n^2 + k$ , whereas ours is a short translation average after all primitive rays have been aggregated. Neither theorem yields a prime number theorem for one fixed quadratic polynomial.

*Remark 5.3* (Context from other short-shift correlations). Higher-order von Mangoldt correlations averaged over one shift variable are studied by Matomäki et al. [10]. Their results provide important methodological context, but are not an input here: the diagonal transfer uses Guth and Maynard [5], and every off-diagonal is bounded by an upper sieve rather than evaluated asymptotically.

## 5.1 Hard Mellin bands

**Corollary 5.4** (Short-shift averaged hard band). *For every  $T \geq 1$ , under the hypotheses of theorem 5.1,*

$$\begin{aligned} \sum_h G(h/H) \int_{-T}^T |\mathcal{R}_{h,D}(X;t)|^2 dt \\ = THXI_0(G)I_2(W) \log X ((\log X)^2 - (\log D)^2) \\ + O\left(THX(\log X)^2 \log \log(3X) + HX^2(\log X)^3\right), \end{aligned} \quad (45)$$

*with constants depending on the fixed data. In particular, if  $T/X \rightarrow \infty$ , the displayed main term is an asymptotic. The same is true for the raw transform.*

*Proof.* Sum (12) over  $h$  with weight  $G(h/H)$ . The main part is  $2T$  times (43). That theorem and (44) also give  $\sum_h G(h/H)\mathcal{E}_{h,D} \ll HX(\log X)^3$ , so the accumulated geometric error in (12) is  $O(HX^2(\log X)^3)$ .  $\square$

## 5.2 Exact Fejér dephasing

**Corollary 5.5** (Short-shift averaged Fejér band). *If  $T\delta_X \geq 1$ , then*

$$\begin{aligned} \sum_h G(h/H) \int_{\mathbb{R}} \kappa(t/T) |\mathcal{R}_{h,D}(X;t)|^2 dt \\ = \frac{1}{2} THXI_0(G)I_2(W) \log X ((\log X)^2 - (\log D)^2) \\ + O\left(THX(\log X)^2 \log \log(3X)\right). \end{aligned} \quad (46)$$

*The same statement holds for the raw transform.*

*Proof.* Use the exact identity (14) and sum theorem 5.1 over the shift.  $\square$

## 6 A parity-conditioned short even-shift ensemble

The positive theorem can be restricted to even shifts without selecting one of them. The correct first-moment model is

$$\mathcal{M}_D^{(2)}(X;t) := 2 \sum_{\substack{n>D, m\geq 1 \\ 2|mn}} u(n)W(mn/X)(n/m)^{it}. \quad (47)$$

For even  $h$ , define

$$\eta_r^{(2)}(h) = \Lambda(r+h) - 2\mathbf{1}_{2 \nmid r}, \quad \mathcal{R}_{h,D}^{(2)} = \mathcal{T}_{h,D} - \mathcal{M}_D^{(2)}. \quad (48)$$

Its factor-ray coefficient is

$$c_{a,b}^{(2)}(h; X, D) = \sum_{\substack{k \geq 1 \\ ak > D}} u(ak) \eta_{abk^2}^{(2)}(h) W(abk^2/X), \quad (49)$$

and  $\mathcal{E}_{h,D}^{(2)}$  denotes the corresponding squared ray energy.

**Proposition 6.1** (Short even-shift first moment). *Under (37), uniformly in  $t \in \mathbb{R}$  and for every  $A > 0$ ,*

$$\frac{2}{H} \sum_{h \equiv 0 \pmod{2}} G(h/H) \mathcal{T}_{h,D}(X; t) = I_0(G) \mathcal{M}_D^{(2)}(X; t) + O_{A,\varepsilon,W,G}(X(\log X)^{-A}). \quad (50)$$

*Proof.* If  $r$  is odd, then every prime target  $r+h \asymp X$  has  $h$  even, and the smooth sum over even  $h$  is therefore the full prime sum from (30), apart from the irrelevant prime 2. It equals  $HI_0(G) + O_A(H(\log X)^{-A})$  outside the same exceptional set. If  $r$  is even, the target is even and its von Mangoldt weight is supported on powers of two, giving a negligible contribution. Group by  $r = mn$  and use the signed exceptional transfer exactly as in the proof of theorem 4.1. The normalization  $2/H$  produces the factor two in (47).  $\square$

**Proposition 6.2** (Short even-shift ray energy). *Let  $G \in C_c^1((0, \infty))$  be nonnegative and nonzero. Under (36)–(37),*

$$\begin{aligned} \sum_{h \equiv 0 \pmod{2}} G(h/H) \mathcal{E}_{h,D}^{(2)}(X) &= \frac{1}{4} H X I_0(G) I_2(W) \log X \\ &\quad \times ((\log X)^2 - (\log D)^2) \\ &\quad + O_{\varepsilon,\eta,W,G}(H X (\log X)^2 \log \log(3X)). \end{aligned} \quad (51)$$

*Proof.* For  $r \in [\alpha X, \beta X]$ , the argument above and the transferred target-square estimate give

$$\sum_{h \equiv 0 \pmod{2}} G(h/H) |\eta_r^{(2)}(h)|^2 = H I_0(G) \log X \mathbf{1}_{2 \nmid r} + O(H) \quad (52)$$

outside a stretched-exponentially sparse set. The weighted exceptional transfer removes that set after summation. The source diagonal on odd products is

$$\begin{aligned} &\sum_{\substack{n > D, m \geq 1 \\ 2 \nmid mn}} u(n)^2 |W(mn/X)|^2 \\ &= \frac{1}{4} X I_2(W) ((\log X)^2 - (\log D)^2) + O_W(X \log X). \end{aligned} \quad (53)$$

Indeed, the odd  $m$  sum has half the continuous density. Restricting  $n$  to odd integers leaves the quadratic logarithmic term in the harmonic  $u(n)^2$  sum unchanged and alters it by only  $O(\log X)$ ; the prime 2 itself contributes a bounded amount.

For distinct target positions, two odd rows are handled by writing  $h = 2j$  and applying the two-linear-forms upper sieve. If either row is even, its target weight is supported on powers of two, and the resulting term is smaller. Thus the even-shift analogue of (42) is

$$\sum_{h \equiv 0 \pmod{2}} G(h/H) |\eta_r^{(2)}(h) \eta_{r'}^{(2)}(h)| \ll_{\varepsilon,W,G} H \log \log(3X).$$

Now expand the squared ray coefficients, use (52)–(53) on the diagonal and (25) off it.  $\square$

The hard- and Fejér-band corollaries follow from (12) and (14), with every leading coefficient exactly one half of its all-shift counterpart.

*Remark 6.3* (No prescribed even shift). The support of  $G(h/H)$  moves to infinity. The proposition gives an asymptotic for a parity-conditioned ensemble; it neither contains nor selects any fixed even shift. In particular, no conclusion about a specified prime gap follows.

## 7 Fixed-shift obstructions inside one ray

The theorem above averages shifts in a window that grows and moves to infinity. This section proves that the fixed-shift problem is not obtained by formally decreasing  $H$ . The first obstruction is completely elementary and uses the actual von Mangoldt coefficients.

Put

$$J_W = \int_0^\infty W(s^2) ds = \frac{1}{2} \int_0^\infty W(v) v^{-1/2} dv. \quad (54)$$

**Theorem 7.1** (A sign-coherent local-obstruction family). *Fix an integer  $h_0 \geq 2$ , assume  $W \geq 0$  and  $W \not\equiv 0$ , and let  $(a, b) = 1$  satisfy*

$$6h_0^2 \mid a, \quad A := ab = o(X). \quad (55)$$

*Suppose the source cutoff is inactive on the radial support; the following explicit sufficient condition will be used:*

$$\frac{D}{a} < \sqrt{\frac{\alpha X}{A}}. \quad (56)$$

*Then*

$$c_{a,b}(h_0; X, D) = \sum_{k \geq 1} W(Ak^2/X) = \sqrt{\frac{X}{A}} J_W + O_W(1). \quad (57)$$

*In particular, this coefficient is not  $o(\sqrt{X/(ab)})$ .*

*Proof.* Every  $ak$  is divisible by both 2 and 3, hence is not a prime power. Thus  $\Lambda(ak) = 0$  and  $u(ak) = -1$ . Also

$$Ak^2 + h_0 = h_0 \left( \frac{A}{h_0} k^2 + 1 \right). \quad (58)$$

Because  $h_0^2 \mid A$ , the factor in parentheses is greater than one and coprime to  $h_0$ . The product in (58) is therefore not a prime power. Consequently  $\eta(Ak^2 + h_0) = -1$ . The two negative factors multiply to one on every supported radial index, and (56) ensures that none is removed.

Finally, Euler summation after the scaling  $k = \sqrt{X/A} s$  gives

$$\sum_k W(Ak^2/X) = \sqrt{X/A} \int_0^\infty W(s^2) ds + O_W(1).$$

Since  $W$  is nonnegative and nonzero,  $J_W > 0$ . □

The family is not confined to a single thin direction. Taking  $a$  to be a multiple of  $6h_0^2$ ,  $b = a + 1$ , and  $a = o(\sqrt{X})$  produces primitive balanced rays whenever (56) holds; this includes  $D \leq c_W \sqrt{X}$  for a sufficiently small fixed  $c_W > 0$ . These balanced examples are not asserted uniformly for larger cutoffs. By contrast, throughout the positive theorem's full range  $D \leq X^{1-\eta}$ , one may take  $b = 1$  and choose  $a$  as a sufficiently large multiple of  $6h_0^2$  with

$$a \asymp_{h_0, W} \max(1, D^2/X).$$

Then (56) holds,  $a = o(X)$ , and the radial length is

$$\sqrt{X/a} \asymp \min(\sqrt{X}, X/D) \geq X^{\min(\eta, 1/2)}.$$

Thus a uniform fixed-shift cancellation theorem must first replace the global center 1 by an appropriate local model or exclude locally obstructed rays. This does not contradict an averaged energy theorem.

### 7.1 An admissible quadratic-prime embedding

Eliminating the preceding fixed-divisor obstruction does not make the fixed-ray problem elementary.

**Theorem 7.2** (Exact quadratic-prime ray). *Take  $(a, b, h) = (6, 1, 1)$  and assume*

$$D < \sqrt{6\alpha X}. \quad (59)$$

*Then*

$$\begin{aligned} c_{6,1}(1; X, D) &= - \sum_{k \geq 1} (\Lambda(6k^2 + 1) - 1) W(6k^2/X) \\ &= N_W(X) - P_W(X), \end{aligned} \quad (60)$$

*where*

$$N_W(X) = \sum_{k \geq 1} W(6k^2/X) = \sqrt{X/6} J_W + O_W(1), \quad (61)$$

$$P_W(X) = \sum_{k \geq 1} \Lambda(6k^2 + 1) W(6k^2/X). \quad (62)$$

*Consequently, for every explicit model  $M(X)$ ,*

$$c_{6,1}(1; X, D) = M(X) + o(\sqrt{X}) \quad (63)$$

*if and only if*

$$P_W(X) = N_W(X) - M(X) + o(\sqrt{X}). \quad (64)$$

*Proof.* The integer  $6k$  has at least two distinct prime factors, so  $\Lambda(6k) = 0$  and  $u(6k) = -1$ . Condition (59) makes the source cutoff inactive on the support. Substitution in (4) proves the first identity, and Euler summation proves (61). The equivalence is then pure algebra.  $\square$

The polynomial  $6k^2 + 1$  is irreducible and has no fixed prime divisor. Proper prime powers do not change the natural-scale interpretation of  $P_W$ . If  $6k^2 + 1$  is an even power of a prime, it lies among solutions of a Pell equation and there are  $O(\log X)$  possibilities in the support. For odd exponents  $j \geq 3$ , the crude bound  $p \ll X^{1/j}$  gives  $O(X^{1/3} \log X)$  possibilities in total. Their aggregate von Mangoldt weight is  $O(X^{1/3} (\log X)^2) = o(\sqrt{X})$ . Thus (64) is a weighted prime-number theorem for this fixed quadratic polynomial at its natural scale. Bateman–Horn predicts it [3]; it is not known. Iwaniec’s theorem that  $n^2 + 1$  assumes infinitely many values with at most two prime factors does not provide such a prime asymptotic [8].

### 7.2 Exact radial blindness of one Mellin coordinate

The arithmetic examples above concern the actual coefficients. There is also a coefficient-independent information boundary. Fix one primitive ray with frequency  $\lambda = \log(a/b)$  and a finite radial set  $K$ . For  $x = (x_k)_{k \in K}$ , all one-Mellin data on that ray are

$$F_x(t) = e^{it\lambda} \sum_{k \in K} x_k. \quad (65)$$

**Proposition 7.3** (Radial-blindness minimax formula). *Let  $K$  be nonempty, let  $B \geq 0$ , let  $1 \leq p \leq \infty$ , let  $q$  be its conjugate exponent, and let  $v \in \mathbb{R}^K$ . Among all, possibly nonlinear, real-valued algorithms using the full signal  $F_x(t)$  for every  $t \in \mathbb{R}$ , the minimax error for recovering  $L_v(x) = \sum_k v_k x_k$  on the ball  $\|x\|_p \leq B$  is*

$$\inf_{\mathcal{A}} \sup_{\|x\|_p \leq B} |L_v(x) - \mathcal{A}(F_x)| = B \operatorname{dist}_{\ell^q}(v, \operatorname{span}\{\mathbf{1}\}). \quad (66)$$

For  $p = 1$ , this equals

$$\frac{B}{2} \left( \max_{k \in K} v_k - \min_{k \in K} v_k \right). \quad (67)$$

*Proof.* For every constant  $\gamma$ , the estimator  $\mathcal{A}(F_x) = \gamma \sum_k x_k$  has worst-case error at most  $B\|v - \gamma \mathbf{1}\|_q$  by Hölder's inequality. Minimize over  $\gamma$ .

Conversely, if  $z \in \ker(\sum_k)$ , the two inputs  $z$  and  $-z$  have the same zero signal. At least one of their recovery errors is  $|L_v(z)|$ . Duality on the kernel gives

$$\sup_{\substack{\|z\|_p \leq B \\ \sum z_k = 0}} |L_v(z)| = B \operatorname{dist}_{\ell^q}(v, \operatorname{span}\{\mathbf{1}\}),$$

which proves the lower bound. For  $q = \infty$ , the best constant is the midrange of  $v$ , giving (67).  $\square$

Increasing the Mellin bandwidth, or retaining any quadratic statistic built from the one-Mellin transform (including the determinant-layer statistics of TPC-11), does not change this minimax value: all such data depend only on the compressed coefficient  $\sum_k x_k$ . The proposition does not say that arithmetic input is useless. It says precisely that frequency resolution in the existing coordinate cannot by itself recover the radial indices.

### 7.3 The endpoint remains a prime-pair boundary

For comparison, define the cutoff-free endpoint energy

$$\mathcal{E}_h^{\text{end}}(X) = \sum_n u(n)^2 \eta(n+h)^2 W(n/X)^2. \quad (68)$$

In the positive range  $D \leq X^{1-\eta}$  this is also the  $m = 1$  part of the fully radial diagonal energy for all large  $X$ , because the support condition forces  $n \asymp_W X > D$ . For every fixed nonzero  $h$ , the Selberg upper sieve and removal of proper prime powers give

$$\begin{aligned} \mathcal{E}_h^{\text{end}}(X) &= (\log X)^2 \sum_n \Lambda(n) \Lambda(n+h) W(n/X)^2 \\ &\quad + O_{h,W}(X \log X). \end{aligned} \quad (69)$$

Indeed, writing  $L = \Lambda(n)$  and  $M = \Lambda(n+h)$  gives

$$(L-1)^2(M-1)^2 = L^2 M^2 - 2L^2 M - 2LM^2 + 4LM + L^2 + M^2 - 2L - 2M + 1.$$

The terms in which either target is a proper prime power contribute  $O_{h,W}(X^{1/2}(\log X)^5)$ , which is admissible. On the remaining prime pairs,

$$L^2 M^2 = ((\log X)^2 + O_{h,W}(\log X)) LM.$$

The two-linear-forms upper sieve gives  $\sum_n LMW(n/X)^2 \ll_{h,W} X$ ; the mixed cubic terms are therefore  $O_{h,W}(X \log X)$ . The prime-square and lower one-variable terms obey  $\sum_{n \asymp X} \Lambda(n)^2 \ll X \log X$ , and all remaining terms are smaller. This proves (69).

For an admissible even  $h$ , an  $X \log^2 X$  asymptotic for (68) with a specified leading constant is therefore equivalent to the corresponding weighted Hardy–Littlewood asymptotic. This is a reduction, not a proof of either assertion. Together with [theorems 7.1](#) and [7.2](#), it shows why the growing-window theorem has no legitimate specialization to  $H = O(1)$ .

## 8 Finite certificate and reproducibility

The analytic asymptotic rests on the theorem of Guth and Maynard [5], the weighted transfer proved here, and the Selberg upper sieve. A finite computation cannot verify those asymptotic inputs. The included deterministic certificate instead checks the exact algebra and the explicit obstructions on which the interpretation rests.

The script `experiments/short_shift_certificate.py` performs the following checks.

- (i) It enumerates a finite factor box, groups it by the product center  $r = mn$ , and verifies that the diagonal source mass equals  $\sum_r A_D(r)$ .
- (ii) It groups the same box by primitive rays and compares the direct Mellin transform with the compressed ray transform at several frequencies.
- (iii) For a synthetic exceptional subset, it records the actual  $A_D$ -mass and verifies the finite Cauchy–Schwarz transfer bound  $\sum_{r \in E} A_D(r) \leq |E|^{1/2} (\sum_r A_D(r)^2)^{1/2}$ .
- (iv) It verifies term by term the sign-coherent identity in [theorem 7.1](#) and the exact quadratic identity in [theorem 7.2](#).
- (v) It checks the Euclidean instance of the radial minimax formula by projecting a test vector onto the constant direction and its orthogonal complement.

The companion unit tests repeat the checks at independent parameter values. Exact integer arithmetic is used for factorization, ray labels, and divisibility; floating-point tolerances occur only in logarithms and Fourier phases. Reproduction requires only the Python standard library:

```
python experiments/short_shift_certificate.py
python -m unittest experiments.test_short_shift_certificate
```

The saved JSON file is a regression certificate for these finite identities. It is not numerical evidence for a fixed-shift prime or prime-pair asymptotic.

## 9 Conclusion and next analytic interface

TPC-11 separated primitive factor rays at fixed shift but used a shift window of length at least  $X$  to evaluate the squared ray coefficients. The present paper fills the range

$$X^{2/15+\varepsilon} \leq H \leq X.$$

The exponent comes from the almost-all short-interval prime number theorem of Guth and Maynard [5]. The factor-ray contribution is the weighted exceptional-set transfer: the divisor-square ledger proves that the sparse exceptional centers cannot carry significant diagonal source mass. Once this issue is closed, the same-ray off-diagonal remains within the reach of a two-linear-forms upper sieve. The resulting energy and Mellin-band formulas are unconditional and retain explicit errors.

The gain is substantial but its quantifiers remain those of a growing ensemble. Any fixed finite collection of shifts has relative mass  $O(1/H)$  and may remain exceptional forever. More strongly, the actual coefficients exhibit sign-coherent locally obstructed rays for every fixed  $h_0 \geq 2$ . After that obstruction is removed, one admissible ray already embeds the open weighted prime number theorem for  $6k^2 + 1$ . The radial minimax identity shows why no increase in the existing Mellin bandwidth can recover the missing  $k$  coordinate without additional structure.

Uniform shifted-divisor technology such as that of Assing et al. [1] controls divisor-weighted shadows of the ray multiplicity. It does not evaluate the  $\Lambda^2$ -weighted large-prime-divisor

component that appears when a fixed ray is opened. Likewise, the quadratic averages of Baier and Zhao [2] and the almost-prime theorem of Iwaniec [8] do not supply a fixed quadratic-prime asymptotic.

The next honest spectral target is therefore a radial completion rather than an extrapolation to  $H = 1$ . On a sector  $ab \asymp R$ , the original ray-frequency spacing is of order  $R^{-1}$ , while the radial length is of order  $\sqrt{X/R}$ . A second Mellin coordinate attached to  $k = (m, n)$  would test the geometric resolution pair

$$T_{\text{ray}} \asymp R, \quad T_{\text{radial}} \asymp \sqrt{X/R}, \quad T_{\text{ray}} T_{\text{radial}}^2 \asymp X.$$

That lift can separate radial indices, but evaluating its fixed-shift diagonal still requires new arithmetic input. The sign-coherent, quadratic-prime, and endpoint reductions in section 7 provide the tests against which any such claim must be measured.

No fixed-shift prime-pair asymptotic, twin-prime lower bound, prime number theorem for a fixed quadratic polynomial, new level of distribution, or violation of sieve parity has been obtained.

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